

Luxury CNC Woodwork Store — Complete Technical Documentation

1. Project Vision

Business Overview

The business is a premium CNC woodwork and furniture manufacturing brand specializing in:

- Luxury wooden beds
- CNC engraved furniture
- Decorative wooden wall panels
- Custom carved wooden products
- Interior woodwork
- Premium handcrafted furniture

The website must position the brand as:

- Premium
- Artistic
- High-end
- Luxury-focused
- Modern
- Trustworthy
- Craftsmanship-driven

This is NOT a normal furniture website.

The website should feel cinematic and emotionally persuasive. It must sell craftsmanship, detail, luxury, and trust.

2. Core Business Goals

Main Objectives

The platform should:

- Showcase CNC craftsmanship.
- Generate leads.
- Allow online custom orders.
- Support direct inquiries.

- Handle large furniture catalogs.
 - Build luxury brand identity.
 - Enable future international growth.
 - Support mobile-first users.
 - Create premium visual storytelling.
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3. Business Requirements Analysis

Extracted Non-Functional Requirements

Performance

- Fast-loading pages.
- Mobile-first optimization.
- Smooth browsing experience.
- Heavy visual content optimization.

Scalability

- Future international expansion.
- Growing product catalog.
- Future multi-warehouse possibility.

Availability

- 24/7 uptime.
- High reliability.
- Backup systems.

Security

- Secure customer information.
- Secure payment integration.
- Admin-only management.
- Audit logging.

Usability

- Non-technical admin panel.
 - Dual-language support.
 - Mobile-first experience.
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4. Recommended Technology Stack

Frontend Stack

Technology	Purpose
Next.js 15	Full-stack React framework
React 19	UI rendering
TypeScript	Type safety
Tailwind CSS	Styling
Framer Motion	Luxury animations
Lenis	Smooth scrolling
GSAP	Cinematic transitions
Shadcn/UI	Premium components
Zustand	State management
React Hook Form	Form management
Zod	Validation

Backend Stack

Technology	Purpose
Node.js	Runtime
Express.js / Next.js APIs	Backend APIs
PostgreSQL	Database
Prisma ORM	Database management
Redis	Caching
BullMQ	Background jobs
JWT	Authentication
bcrypt	Password hashing

Infrastructure Stack

Technology	Purpose
Vercel	Frontend hosting
Railway / AWS	Backend hosting
Cloudflare	CDN + DDoS protection
Neon / Supabase	Managed PostgreSQL
Upstash Redis	Managed Redis
GitHub	Source control
GitHub Copilot	AI-assisted development
GitHub Actions	CI/CD
Cloudinary	Image optimization
Sentry	Error monitoring
PostHog	Analytics

5. Website Design Philosophy

Luxury Design Direction

The website should feel:

- Minimal
- Cinematic
- Elegant
- Architectural
- Premium
- Spacious
- Sophisticated

References:

- Apple
- Restoration Hardware
- Aesop
- Poliform
- B&B Italia
- Bentley Home

- Luxury interior design studios
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6. UI/UX Strategy

Core UX Principle

Users are not buying wood.

They are buying:

- craftsmanship
- prestige
- luxury
- aesthetics
- identity
- interior experience

The UI must emotionally communicate craftsmanship.

7. Animation System

Primary Animation Stack

Use:

- Framer Motion
 - Lenis
 - GSAP (selectively)
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Recommended Animation Style

Use:

- slow transitions
- cinematic fades
- scroll storytelling
- image reveals
- parallax depth
- smooth product transitions
- elegant hover effects
- luxury pacing

Avoid:

- cheap flashy effects
 - excessive motion
 - fast aggressive transitions
 - overdone parallax
-

8. Homepage Structure

Recommended Sections

Hero Section

- Fullscreen cinematic video
 - CNC cutting visuals
 - Wood craftsmanship shots
 - Luxury typography
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Featured Collections

- Luxury beds
 - Wall panels
 - Decorative woodwork
 - CNC carvings
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Craftsmanship Section

Show:

- CNC machines
 - carving process
 - finishing process
 - material quality
-

Testimonials

- customer interiors
 - completed projects
 - client reviews
-

Gallery Showcase

High-resolution cinematic project showcase.

CTA Section

- WhatsApp consultation
 - custom quote request
 - showroom inquiry
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9. Core Pages

Public Pages

- Home
 - Collections
 - Product Details
 - Custom Orders
 - Gallery
 - About
 - Contact
 - CNC Process
 - FAQs
 - Blog
 - Cart
 - Checkout
 - Returns Policy
 - Privacy Policy
-

Admin Pages

- Dashboard
 - Products
 - Orders
 - Inventory
 - Customers
 - Analytics
 - Quotations
 - Custom Requests
 - Content Management
 - Shipment Management
-

10. Database Design

Core Tables

Users

```
id
name
email
phone
password_hash
role
created_at
```

Products

```
id
name
slug
description
price
category_id
material
finish
created_at
```

ProductImages

```
id
product_id
image_url
alt_text
```

CustomOrders

```
id
user_id
design_reference
```



```
requirements
budget
status
created_at
```

Orders

```
id
user_id
total_amount
status
payment_status
shipping_status
created_at
```

AuditLogs

```
id
admin_id
action
entity
created_at
```

11. Product System

Required Features

Product Features

- Multiple images
 - Product variants
 - Material selection
 - Size selection
 - Finish selection
 - Custom engraving requests
 - Dynamic pricing
-

Product Categories

Examples:

- CNC Beds
 - Wall Panels
 - Wardrobes
 - Decorative Pieces
 - Dining Tables
 - Interior Panels
-

12. Custom Order System

Critical Feature

This business depends heavily on custom requests.

Therefore the website must include:

Custom Quote System

Users should:

- upload inspiration images
 - enter dimensions
 - choose wood type
 - describe design
 - request pricing
-

Recommended Workflow

```
graph TD; A[Customer Request] --> B[Admin Review]; B --> C[Quotation]; C --> D[Approval]; D --> E[Production];
```

↓
Delivery

13. Authentication & Authorization

Recommended Authentication

Use:

- JWT access tokens
 - refresh tokens
 - secure cookies
-

Security Features

- bcrypt hashing
 - role-based access
 - admin protection
 - secure sessions
 - CSRF protection
 - XSS prevention
-

14. Payment Integration

Recommended Payment Systems

Pakistan

- JazzCash
- Easypaisa
- Bank Transfer

Future International

- Stripe
 - PayPal
-

15. JazzCash Integration

Recommended Architecture

```
graph TD; Frontend --> Backend_API[Backend API]; Backend_API --> JazzCash_API[JazzCash API]; JazzCash_API --> Webhook_Verification[Webhook Verification]; Webhook_Verification --> Database_Update[Database Update];
```

Required Credentials

- Merchant ID
- Password
- Integrity Salt
- Return URL

Security Rules

NEVER:

- store card data
- expose API secrets
- trust frontend payment responses

ALWAYS:

- verify payments server-side
- use HTTPS
- validate callbacks

16. Media & Image System

Critical Requirement

Furniture websites live or die based on visuals.

Bad visuals destroy perceived luxury.

Recommended Image System

Use:

- Cloudinary

Why?

- image optimization
 - automatic compression
 - responsive delivery
 - CDN support
 - WebP generation
-

Media Standards

Images

- 4K photography
- WebP format
- optimized delivery
- cinematic lighting

Videos

- CNC cutting footage
 - slow-motion craftsmanship shots
 - luxury interior showcase
-

17. Performance Engineering

Performance Targets

Metric	Goal
Homepage	<2 seconds
Product Pages	<2.5 seconds
Mobile Lighthouse	90+
CLS	Minimal

Optimization Strategy

Frontend

- lazy loading
- route prefetching
- image optimization
- code splitting
- font optimization

Backend

- Redis caching
 - optimized queries
 - pagination
 - CDN delivery
-

18. SEO Strategy

SEO Importance

Luxury furniture businesses rely heavily on:

- Google traffic
- Pinterest traffic
- Instagram discovery
- visual search

Ignoring SEO is stupidity.

SEO Requirements

- server-side rendering
 - schema markup
 - product schema
 - sitemap.xml
 - clean URLs
 - metadata optimization
 - blog content
 - optimized images
-

19. Mobile-First Strategy

Client Requirement

Primary audience:

- mobile users

Therefore:

Mobile Rules

- thumb-friendly design
 - large buttons
 - optimized scrolling
 - fast loading
 - lightweight interactions
 - responsive typography
-

20. Internationalization

Languages

Required:

- English
 - Urdu
-

Recommended Libraries

Use:

- next-intl OR
 - i18next
-

21. Inventory Management

Required Features

- stock tracking
 - raw material tracking
 - custom order tracking
 - SKU management
 - inventory alerts
-

22. Shipping System

Required Features

- domestic shipping
 - international shipping
 - tracking numbers
 - shipment statuses
 - shipping zones
-

Recommended Integrations

Pakistan:

- TCS
- Leopards Courier
- Trax

International:

- DHL
- FedEx

23. Admin Dashboard

Critical Requirement

Client requested:

- non-technical usability

Therefore admin UX must be extremely simple.

Dashboard Features

- drag-and-drop image upload
 - simple product editing
 - order tracking
 - quotation management
 - inventory control
 - analytics dashboard
-

24. Analytics System

Recommended Stack

Tool	Purpose
Google Analytics	Traffic analytics
PostHog	Product analytics
Meta Pixel	Facebook ads
TikTok Pixel	TikTok ads

Track Metrics

- conversion rate
- inquiry rate
- custom quote submissions
- product views
- mobile engagement
- top-performing products

25. Security Engineering

Required Security Measures

Backend Security

- rate limiting
- input validation
- SQL injection prevention
- CSRF protection
- XSS prevention
- secure headers

Recommended Security Stack

Tool	Purpose
Helmet.js	Security headers
bcrypt	Password hashing
Zod	Validation
Cloudflare	DDoS protection

26. Backup & Recovery

Backup Strategy

Daily

- full database backup

Hourly

- incremental backup

Weekly

- offsite backup
-

Storage

Store backups:

- cloud storage
 - secondary region
-

27. Scalability Engineering

Future Scalability Goals

System should support:

- traffic spikes
 - international traffic
 - large media galleries
 - growing product catalog
 - future showroom expansion
-

Scalability Strategy

Frontend

- edge caching
- CDN optimization

Backend

- stateless APIs
- horizontal scaling
- Redis caching

Database

- indexing
 - read replicas
 - optimized queries
-

28. Deployment Architecture

Recommended Setup

Frontend

Deploy on:

Vercel

Backend

Deploy on:

Railway / AWS

Database

Deploy on:

Neon / Supabase

29. CI/CD Pipeline

Recommended Workflow

```
graph TD; A[GitHub Push] --> B[GitHub Actions]; B --> C[Testing]; C --> D[Build Verification]; D --> E[Deployment];
```

30. GitHub Copilot Strategy

Best Use Cases

Use Copilot for:

- boilerplate
 - APIs
 - CRUD systems
 - forms
 - components
 - repetitive logic
-

Major Warning

Copilot produces:

- insecure code
- bloated logic
- terrible architecture
- performance issues

Review everything manually.

31. Testing Strategy

Required Testing

Unit Testing

Use:

Vitest

E2E Testing

Use:

Playwright

API Testing

Use:

Postman

32. Monitoring & Observability

Recommended Stack

Tool	Purpose
Sentry	Error monitoring
UptimeRobot	Uptime monitoring
PostHog	Analytics
Logtail	Logging

33. Recommended Development Roadmap

Week 1

- architecture setup
- database schema
- design system
- project structure

Week 2

- homepage
- collections
- product pages

Week 3

- authentication
- cart
- checkout
- quotation system

Week 4

- JazzCash integration
- admin dashboard
- analytics

Week 5

- SEO optimization
 - testing
 - deployment
 - performance tuning
-

34. Design System Recommendations

Typography

Use:

- Geist
 - Inter
 - Satoshi
-

Color Direction

Use:

- warm neutrals
- deep charcoal
- wood-inspired palettes
- luxury monochrome

Avoid:

- oversaturated colors
 - cheap gradients
 - random neon palettes
-

35. Common Mistakes To Avoid

Critical Mistakes

1. Over-animation

Luxury brands move slowly and elegantly. Not like TikTok garbage.

2. Bad product photography

Weak visuals destroy trust instantly.

3. Heavy frontend

Cinematic does NOT mean slow.

4. Weak mobile optimization

Most Pakistani traffic is mobile.

5. Poor inquiry system

Custom order flow is core business.

6. Ignoring SEO

Furniture discovery heavily depends on search traffic.

7. Cheap UI

Luxury customers judge quality within seconds.

36. Final Technical Recommendation

Best Stack For This Project

Frontend

- Next.js
- TypeScript
- Tailwind CSS
- Framer Motion
- Lenis
- GSAP

- Shadcn/UI

Backend

- Node.js
- Prisma
- PostgreSQL
- Redis

Infrastructure

- Vercel
- Cloudflare
- Neon
- GitHub Actions

Payments

- JazzCash
- Easypaisa
- Stripe (future)

37. Final Strategic Advice

The biggest beginner mistake is thinking luxury comes from excessive animation.

It does not.

Luxury comes from:

- typography
- spacing
- responsiveness
- image quality
- performance
- consistency
- restrained motion
- visual hierarchy

A slow cinematic website is garbage.

If performance dies, the luxury perception dies with it.

The real challenge is creating:

- cinematic visuals WITH

- elite performance.

That balance is what separates amateur portfolios from real production-grade luxury websites.

38. Final Conclusion

This project is not a simple furniture catalog.

It is:

- brand engineering
- UX psychology
- conversion optimization
- performance engineering
- infrastructure engineering
- visual storytelling
- luxury perception design

The website must make users emotionally feel:

“This craftsmanship is premium.”

That is the real objective.